Chapter 2. Supervised Learning and Model Assessment

Lecture 4 (regression function, parametric vs nonparametric, KNN, bias–variance, adjusted R^2, AIC, BIC)

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1 The Supervised Learning Setup

Motivation. We observe pairs (X, Y) and want to predict Y for new X. Writing $Y = f(X) + \epsilon$ separates what can be learned (f) from what cannot (ϵ), and this split explains why even a perfect method has nonzero test error.

Example: advertising data

A marketing data set records product `Sales` and the advertising budgets `TV`, `Radio`, `Digital` in many markets. Goal: learn f with $\texttt{Sales} \approx f(\texttt{TV}, \texttt{Radio}, \texttt{Digital})$. Response $Y = \texttt{Sales}$; predictors $X = (X_1, X_2, X_3)$; model

$$Y = f(X) + \varepsilon,$$

with f the unknown regression function and ε noise or unobserved factors ($\mathbb{E}\varepsilon = 0$, independent of X). A second example: income of 30 individuals versus years of education, seniority and marital status — which variables truly explain income?

1.1 Why estimate f?

Prediction: use $\hat{f}$ to predict Y at a new $X = x$. **Inference**: understand how X influences Y — which predictors matter, and the direction and size of their effects (which media increase sales, which channel has the largest impact, what is the expected change in sales from an extra \$1,000 on TV?).

1.2 The regression function

At a given value, say $X = 4$, many Y values are possible. What is the best prediction?

Theorem: the conditional mean is the optimal predictor

$$f(x) = \arg\min_g \mathbb{E}\big[(Y - g(X))^2 \mid X = x\big] = \mathbb{E}(Y \mid X = x),$$

called the **regression function**. Moreover, for any g, $\mathbb{E}[(Y - g(X))^2] = \mathbb{E}[(Y - f(X))^2] + \mathbb{E}[(f(X) - g(X))^2]$.

Proof [Supplement]. For a constant c, $\mathbb{E}[(Y - c)^2 \mid X = x] = \mathrm{Var}(Y \mid X = x) + (f(x) - c)^2$, minimized at $c = f(x)$. For the decomposition, write $Y - g = (Y - f) + (f - g)$; the cross term vanishes because $\mathbb{E}[(Y - f(X)) \mid X] = 0$. ■

The term $\mathrm{Var}(Y \mid X = x) = \mathrm{Var}(\varepsilon)$ is the **irreducible error**: even the true f cannot remove it. The term $\mathbb{E}(f - g)^2$ is the **reducible error** that learning tries to shrink.

2 Parametric Approach

Parametric methods impose structure on f:

1. specify a functional form, e.g. the linear model $f_L(X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$;
2. estimate the parameters $(\beta_0, \ldots, \beta_p)$ from data (e.g. by least squares).

f is restricted to a low-dimensional parameter space: this gains interpretability and stability but risks **model misspecification**. Adding parameters increases flexibility: the linear model $\hat{f}_L(X) = \hat{\beta}_0 + \hat{\beta}_1 X$ versus the quadratic $\hat{f}_Q(X) = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2$; with two inputs (education, seniority) the linear model is a plane and a polynomial model a curved surface.

3 Nonparametric Approach

3.1 Local averaging

We rarely have many observations with exactly $X = 4$, so $\mathbb{E}(Y \mid X = x)$ cannot be computed directly. Instead we average nearby points:

$$\hat{f}(x) = \mathrm{Ave}(Y \mid X \in \mathcal{N}(x)),$$

with $\mathcal{N}(x)$ a neighborhood of x.

Definition: K-nearest neighbors regression

$$\hat{f}(x) = \frac{1}{K} \sum_{i \in \mathcal{N}_K(x)} y_i,$$

where $\mathcal{N}_K(x)$ is the set of the K training points closest to x.

Other nonparametric methods: kernel regression, smoothing splines (Ch. 7), Gaussian-process regression. With $K = 1$ the fit interpolates the data and is very rough; with $K = 9$ it is smoother.

Theorem: bias and variance of KNN at a point

Let $Y = f(X) + \varepsilon$ with $\mathrm{Var}\varepsilon = \sigma^2$ and condition on the inputs. With neighbors $x_{(1)}, \ldots, x_{(K)}$ of x_0,

$$\mathbb{E}\big[(Y_0 - \hat{f}(x_0))^2\big] = \sigma^2 + \Big[f(x_0) - \frac{1}{K}\sum_{l=1}^{K} f(x_{(l)})\Big]^2 + \frac{\sigma^2}{K}.$$

Proof [Supplement]. $\hat{f}(x_0) = \frac{1}{K}\sum_l f(x_{(l)}) + \frac{1}{K}\sum_l \varepsilon_{(l)}$; its variance is σ^2/K and its mean is the average of f over the neighbors. Apply the bias–variance decomposition (§4.3). ■

Small K: low bias (neighbors are close), high variance σ^2/K. Large K: neighbors spread out, bias grows. K is the complexity parameter (*smaller* K = *more* flexible).

3.2 Pros and cons

Nonparametric methods make no explicit assumption about the form of f, giving flexible estimates that try to get close to the data without being too rough or wiggly. The price: they need a very large number of observations (and suffer from the curse of dimensionality, Ch. 1). An extremely flexible fit can reproduce the training data almost perfectly and still predict badly.

4 Assessing Model Accuracy

4.1 Complexity, underfitting, overfitting

Simple models impose strong structural assumptions (parametric, e.g. linear regression); nonparametric models (nearest neighbors, splines) are more flexible. **Underfitting**: the model is too simple,

producing systematic errors. **Overfitting**: the model follows the training data too closely and fits noise rather than signal.

4.2 Training error and test error

For a model fitted on $\mathrm{Tr} = \{(x_i, y_i)\}_{i=1}^{n}$,

$$\mathrm{MSE}_{\mathrm{Tr}} = \mathrm{Ave}_{i \in \mathrm{Tr}} \big[y_i - \hat{f}(x_i)\big]^2, \qquad \mathrm{MSE}_{\mathrm{Te}} = \mathrm{Ave}_{i \in \mathrm{Te}} \big[y_i - \hat{f}(x_i)\big]^2,$$

with $\mathrm{Te} = \{(x_i, y_i)\}_{i=1}^{M}$ fresh observations. Training error is overly optimistic, especially for flexible models; what we care about is performance on new data. Three simulated cases illustrate it:

(1) a moderately curved truth — training MSE falls monotonically with flexibility while test MSE is U-shaped;

(2) a nearly linear truth — the linear model and a smooth fit do best;

(3) a wiggly truth with low noise — very flexible fits do best.

4.3 Bias–variance trade-off

Theorem: decomposition of the expected test error

Let (x_0, y_0) be a new observation, $y_0 = f(x_0) + \varepsilon_0$, $f(x) = \mathbb{E}(Y \mid X = x)$, and let $\hat{f}$ be fitted on an independent training set. Then

$$\mathbb{E}\big[(y_0 - \hat{f}(x_0))^2\big] = \underbrace{\mathrm{Var}(\hat{f}(x_0))}_{\text{variance}} + \underbrace{\big[\mathrm{Bias}(\hat{f}(x_0))\big]^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}(\varepsilon)}_{\text{irreducible}} .$$

Proof [Supplement]. $y_0 - \hat{f} = \varepsilon_0 + (f - \mathbb{E}\hat{f}) + (\mathbb{E}\hat{f} - \hat{f})$; the three terms are uncorrelated (ε_0 is independent of the training data and has mean 0; the second term is constant; the third has mean 0). Square and take expectations. ■

As flexibility increases, bias decreases and variance increases. The challenge is to find a method for which *both* are low.

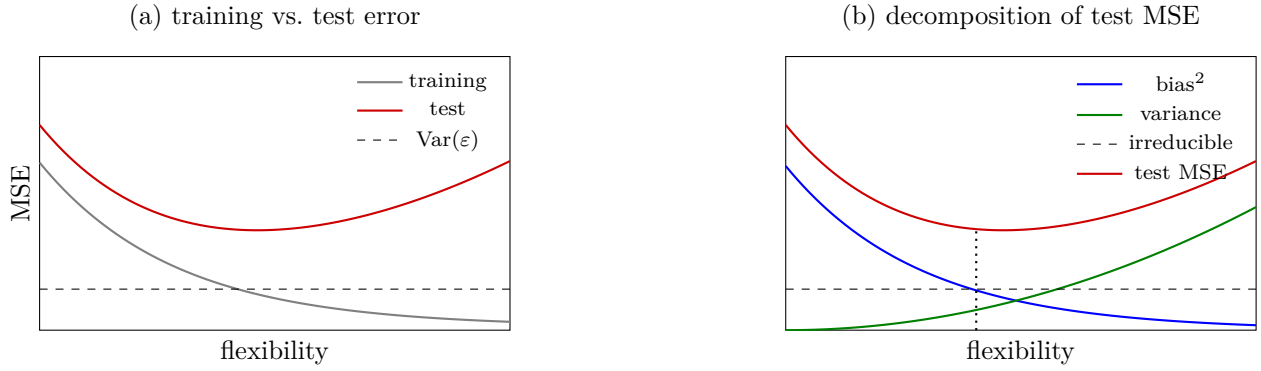
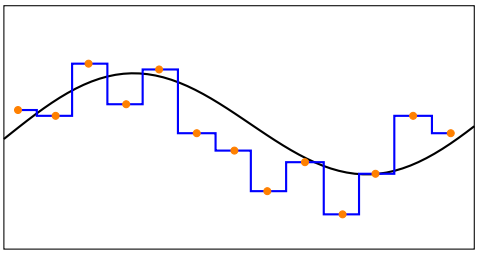


Figure 1: Schematic bias–variance trade-off. Training error always falls with flexibility; test error is U-shaped and never goes below the irreducible error. The dotted line marks the optimal flexibility.

$K = 1$: low bias, high variance



$K = 9$: smoother, more bias

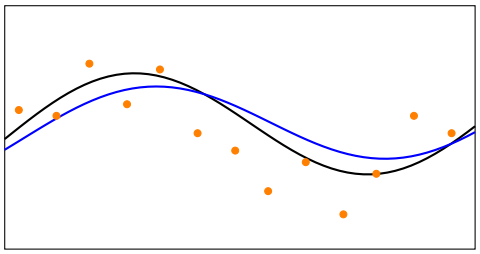


Figure 2: KNN regression in one dimension (schematic). Black: true f; blue: fit. $K = 1$ reproduces every training point (step function through the data); larger K averages more neighbors and flattens peaks and troughs.

5 Criteria Based on the Training Data

Besides test error, some criteria evaluate models on the training data by adding a complexity penalty:

$$\text{Score} = \text{goodness of fit} + \text{complexity penalty}, \qquad \text{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

5.1 Adjusted R^2

$R^2 = 1 - \text{RSS}/\text{TSS}$ always increases when a predictor is added (RSS cannot increase when the model is enlarged). The adjusted version penalizes complexity:

$$R^2_{\text{adj}} = 1 - \frac{\text{RSS}/(n-p-1)}{\text{TSS}/(n-1)}.$$

Adding a predictor increases R^2_{adj} only if it lowers $\text{RSS}/(n-p-1)$, i.e. if it improves the fit sufficiently (equivalently, if its t-statistic exceeds 1 in absolute value).

5.2 AIC and BIC

For a Gaussian linear model with p predictors,

$$\text{AIC} = n \log(\text{RSS}/n) + 2(p+1), \qquad \text{BIC} = n \log(\text{RSS}/n) + \log(n)(p+1);$$

choose the model with the smallest value. In general, with k parameters, $\text{AIC} = -2 \log L(\hat{\theta}) + 2k$ and $\text{BIC} = -2 \log L(\hat{\theta}) + k \log n$. (For the Gaussian model $-2 \log L(\hat{\theta}) = n \log(\text{RSS}/n) + \text{const.}$)

Derivation: information-theoretic interpretation of AIC

Measure model quality by the Kullback–Leibler divergence $\text{KL}(p_0 \| p_\theta) = \mathbb{E}_{p_0} \log \frac{p_0(X)}{p_\theta(X)}$; smaller is closer to the truth. Up to a constant this equals $-\mathbb{E}_{p_0} \log p_\theta(X)$, the expected log-loss on a *new* observation. The in-sample maximized log-likelihood $\log L(\hat{\theta})$ is optimistic because $\hat{\theta}$ was fitted to the same data. A second-order Taylor expansion around the best parameter θ^* (correct model, regular case) shows that each of the two steps — fitting and evaluating on new data — contributes about $k/2$ in expectation, so

$$\mathbb{E}\big[-2\, \mathbb{E}_{\text{new}} \log p_{\hat{\theta}}(X^{\text{new}}) \big] \approx \mathbb{E}\big[-2 \log L(\hat{\theta}) \big] + 2k.$$

Hence $\text{AIC} = -2 \log L(\hat{\theta}) + 2k$ estimates the expected out-of-sample deviance (Akaike).

Derivation: Bayesian interpretation of BIC

For candidate models $M_1, \dots, M_K$ with equal prior weights, Bayesian model selection picks the largest posterior $P(M_k \mid \text{data}) \propto \int L_k(\theta)\pi_k(\theta)d\theta$. A Laplace approximation of the integral around $\hat{\theta}_k$ gives

$$\log P(M_k \mid \text{data}) \approx \log L(\hat{\theta}_k) - \frac{k}{2}\log n + O(1)$$

(Schwarz, 1978). Multiplying by -2 gives BIC, so BIC approximates the Bayesian model evidence and the smallest BIC corresponds to the largest posterior probability.

	Penalty	Goal	Model size	Consistent?
AIC	$2k$	prediction	larger models	no
BIC	$k \log n$	true-model selection	smaller models	yes

AIC focuses on predictive performance; BIC prefers simpler models when n is large ($\log n > 2$ once $n \geq 8$). **Consistency**: if the true model is among the candidates, BIC selects it with probability $\to 1$, whereas AIC keeps a positive probability of overfitting. In practice: AIC (or CV) for prediction, BIC to identify the true structure.

Summary

- The regression function $\mathbb{E}(Y \mid X = x)$ is the optimal predictor under squared loss; $\text{Var}(\varepsilon)$ is irreducible.
- Parametric methods assume a form for f (interpretable, may be misspecified); nonparametric methods such as KNN average locally (flexible, data-hungry).
- Training error is optimistic; test error is U-shaped in flexibility because of the bias–variance trade-off.
- Adjusted R^2, AIC and BIC penalize complexity; AIC targets prediction, BIC consistency.

6 Selected exercises

Motivation. Three derivations that the chapter uses without proof: the optimal predictor under absolute loss, the $|t| > 1$ rule for adjusted R^2, and the Gaussian log-likelihood behind the AIC and BIC formulas.

Exercise 2.1 (Absolute loss and the conditional median)

(a) Let Y be a random variable with $\mathbb{E}|Y| < \infty$ and let m be a median, $\Pr(Y \leq m) \geq \frac{1}{2}$ and $\Pr(Y \geq m) \geq \frac{1}{2}$. Show that $\mathbb{E}|Y - c| \geq \mathbb{E}|Y - m|$ for every $c \in \mathbb{R}$.
(b) Deduce that $\mathbb{E}|Y - g(X)|$ is minimized over functions g by a conditional median $g(x) = \text{median}(Y \mid X = x)$.

Solution [Supplement].

(a) All expectations below are finite because $\mathbb{E}|Y| < \infty$.

Step 1 ($c > m$). If $y \leq m$, then $|y - c| - |y - m| = (c - y) - (m - y) = c - m$. For every y, the triangle inequality gives $|y - c| - |y - m| \geq -(c - m)$. Hence

$$\mathbb{E}[|Y - c| - |Y - m|] \geq (c - m)\Pr(Y \leq m) - (c - m)\Pr(Y > m) = (c - m)\{2\Pr(Y \leq m) - 1\} \geq 0.$$

Step 2 ($c < m$). Symmetrically, $|y-c|-|y-m| = m-c$ for $y \geq m$ and $\geq -(m-c)$ always, so the difference has expectation $\geq (m-c)\{2\Pr(Y \geq m) - 1\} \geq 0$.

(b) Let $m(x)$ be a median of the conditional distribution of Y given $X = x$ (e.g. $m(x) = \inf\{c : \Pr(Y \leq c \mid X = x) \geq \frac{1}{2}\}$, which is measurable). Applying (a) to that conditional distribution, $\mathbb{E}[|Y - g(X)| \,|\, X = x] \geq \mathbb{E}[|Y - m(X)| \,|\, X = x]$ for every x; taking expectations over X gives $\mathbb{E}|Y - g(X)| \geq \mathbb{E}|Y - m(X)|$.

Thus squared loss targets $\mathbb{E}(Y \mid X)$ and absolute loss the conditional median; least absolute deviation regression (Ch. 5) estimates the latter. ■

Exercise 2.2 (Adjusted R^2 rises iff $|t| > 1$)

Let M_0 have design $X_0 \in \mathbb{R}^{n\times(p+1)}$ (intercept and p predictors) and residual sum of squares RSS_0, and let M_1 add one predictor z, with design $X_1 = [X_0, z]$ of full column rank $p+2 < n$ and $\mathrm{RSS}_1 > 0$. Let $t = \hat\gamma/\mathrm{SE}(\hat\gamma)$ be the usual t-statistic of the coefficient $\hat\gamma$ of z in M_1, with $\hat\sigma^2 = \mathrm{RSS}_1/(n-p-2)$.
(a) Show that $\mathrm{RSS}_0 - \mathrm{RSS}_1 = t^2\,\mathrm{RSS}_1/(n-p-2)$.
(b) Show that $R^2_{\mathrm{adj}}(M_1) > R^2_{\mathrm{adj}}(M_0)$ if and only if $|t| > 1$.

Solution [Supplement].

(a) Let H_0 be the orthogonal projection onto $\mathrm{col}(X_0)$ and $\tilde z = (I - H_0)z$; $\tilde z \neq 0$ since X_1 has full column rank.

Step 1. $\mathrm{col}(X_1) = \mathrm{col}(X_0) \oplus \mathrm{span}(\tilde z)$ with $\tilde z \perp \mathrm{col}(X_0)$, so the projection onto $\mathrm{col}(X_1)$ is $H_1 = H_0 + \tilde z\tilde z^\top/(\tilde z^\top \tilde z)$.

Step 2. Since $\tilde z^\top H_0 = 0$, we have $\tilde z^\top y = \tilde z^\top (I - H_0)y$ and

$$\mathrm{RSS}_1 = y^\top (I - H_1)y = \mathrm{RSS}_0 - \frac{(\tilde z^\top y)^2}{\tilde z^\top \tilde z}.$$

Step 3. Put $c = \tilde z^\top y/\tilde z^\top \tilde z$. Then $H_1 y = H_0 y + c\tilde z = (H_0 y - cH_0 z) + cz$, where $H_0 y - cH_0 z \in \mathrm{col}(X_0)$. The coefficients of $H_1 y$ in the basis given by the columns of X_1 are unique, so $\hat\gamma = c$. As a linear function of y, $\mathrm{Var}(\hat\gamma) = \sigma^2/\tilde z^\top \tilde z$, so $\mathrm{SE}(\hat\gamma)^2 = \hat\sigma^2/\tilde z^\top \tilde z$.

Step 4. Hence $t^2 = \dfrac{(\tilde z^\top y)^2}{\tilde z^\top \tilde z\, \hat\sigma^2} = \dfrac{\mathrm{RSS}_0 - \mathrm{RSS}_1}{\hat\sigma^2}$, which is (a).

(b) Both models share TSS, so M_1 has the larger R^2_{adj} iff $\mathrm{RSS}_1/d < \mathrm{RSS}_0/(d+1)$ with $d = n-p-2$. By (a), $\mathrm{RSS}_0 = \mathrm{RSS}_1(1 + t^2/d)$, so the condition reads $d + 1 < d + t^2$, i.e. $t^2 > 1$. ■

Exercise 2.3 (The Gaussian log-likelihood in AIC and BIC)

Let $y = X\beta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2 I_n)$, $X \in \mathbb{R}^{n\times(p+1)}$ of full column rank, and assume $\mathrm{RSS} > 0$ for the least-squares fit.
(a) Show that $\max_{\beta,\sigma^2} \log L(\beta, \sigma^2) = -\frac{n}{2}\{\log(2\pi) + \log(\mathrm{RSS}/n) + 1\}$.
(b) Deduce that counting σ^2 as a parameter ($k = p + 2$) changes AIC and BIC only by a constant common to all models fitted to the same n observations, so the model rankings agree with the formulas of this chapter.

Solution [Supplement].

(a)

Step 1. $\log L = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|y - X\beta\|^2$. For each fixed σ^2 this is maximized by minimizing $\|y - X\beta\|^2$, i.e. at the least-squares $\hat\beta$, giving $h(s) = -\frac{n}{2}\log(2\pi s) - \mathrm{RSS}/(2s)$ with $s = \sigma^2$.

Step 2. $h'(s) = -\frac{n}{2s} + \frac{\mathrm{RSS}}{2s^2} = \frac{\mathrm{RSS}-ns}{2s^2}$ is positive for $s < \mathrm{RSS}/n$ and negative beyond, so h is maximized at $\hat{\sigma}^2 = \mathrm{RSS}/n$, where $h = -\frac{n}{2}\log(2\pi\,\mathrm{RSS}/n) - \frac{n}{2}$.

(b) By (a), $-2\log L(\hat{\theta}) = n\log(\mathrm{RSS}/n) + n(1+\log 2\pi)$. With $k = p+2$,

$$\mathrm{AIC} = n\log(\mathrm{RSS}/n) + 2(p+1) + \{n(1+\log 2\pi) + 2\},$$
$$\mathrm{BIC} = n\log(\mathrm{RSS}/n) + (p+1)\log n + \{n(1+\log 2\pi) + \log n\}.$$

The braces do not depend on the model, so differences between models, and hence the selected model, are unchanged. ■